



Wake Forest University
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HOW TO DEVELOP A SCIENTIFIC ABSTRACT

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An abstract is a **brief summary** of a research article, thesis, review, conference proceeding, or any in-depth analysis of a particular subject and is often used to help the reader **quickly ascertain the paper's purpose.**



WHAT IS AN
ABSTRACT?

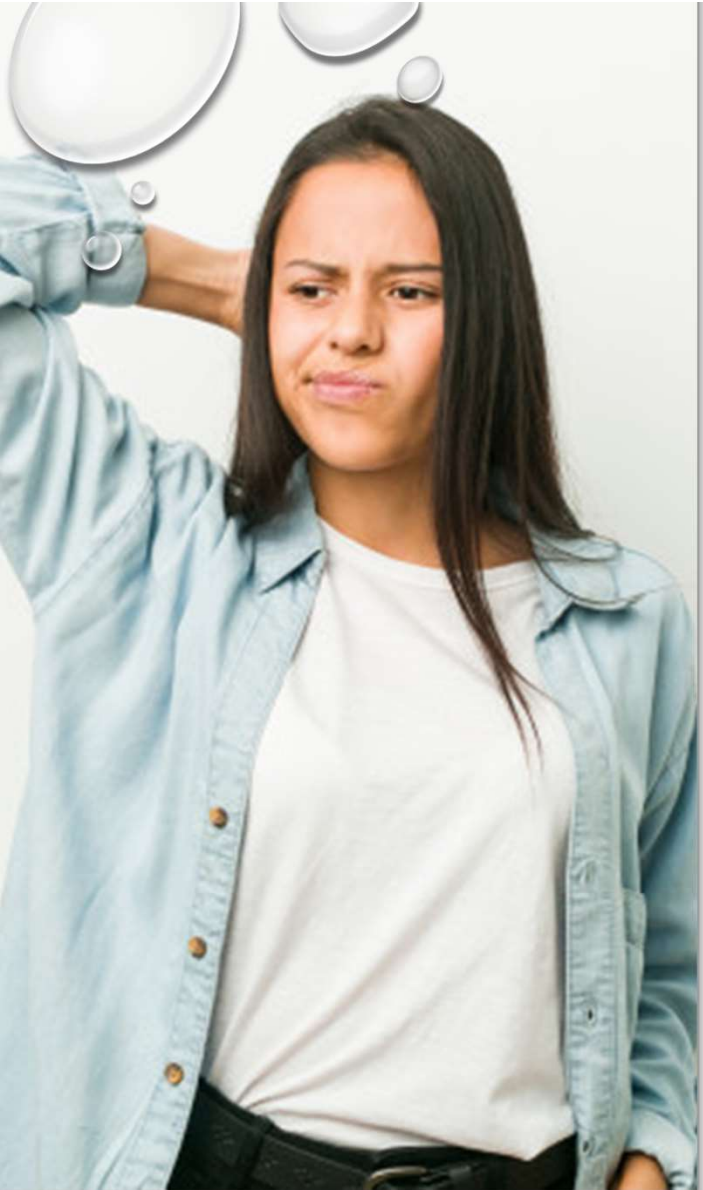


WHO HAS WRITTEN AN ABSTRACT BEFORE?

WHAT IS THE PURPOSE OF A SCIENTIFIC ABSTRACT?

- **INTRODUCE** JOURNAL ARTICLES
- **INFORM** READERS ABOUT THE ARTICLE'S CONTENT
- HELP READERS **DECIDE** IF THEY WANT TO READ THE ARTICLE
- **OVERVIEW** OF RESEARCH STUDIES, CONFERENCE PROGRAMS AND BOOKS





WHY SHOULD I KNOW HOW TO WRITE ABSTRACTS?

- HELPS YOU **PRESENT** COMPLEX INFORMATION IN A CLEAR, CONCISE MANNER
- HELPS YOU **READ** ABSTRACTS MORE EFFECTIVELY
- HELPS YOU **CONDENSE** REPORT INFORMATION INTO A SHORT FORMAT FOR DATABASE SEARCHES
- HELPS YOU **WRITE** ABSTRACTS FOR FUTURE PUBLICATIONS
- HELPS YOU **CONDUCT** RESEARCH



ELEMENTS OF AN EFFECTIVE ABSTRACT

**FOLLOWS
THE
SCIENTIFIC
METHOD**

**CAN STAND
ALONE**

**INTELLIGIBLE
TO A WIDE
AUDIENCE**

SCIENTIFIC ABSTRACTS USE AN “INTRODUCTION-BODY-CONCLUSION” STRUCTURE

INTRODUCTION

Orient the reader

Identify the focus/purpose

Outline scope

State thesis

BODY

TOPIC SENTENCE 1

SUPPORTING DETAILS

CONCLUDING SENTENCE 1

TOPIC SENTENCE 2

SUPPORTING DETAILS

CONCLUDING SENTENCE 2

ETC.

CONCLUSION

RESTATE THESIS

SUMMARIZE ARGUMENT

SCIENTIFIC ABSTRACTS USE AN “INTRODUCTION – BODY-CONCLUSION” STRUCTURE



Purpose (Background)



Research Question(s)



Methods



Findings



Conclusions



Introduction

Body

Conclusion

THERE MAY BE SPECIFIC REQUIREMENTS REGARDING THE FORMAT OF AN ABSTRACT

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Clin Toxicol (Phila). 2014 Jun;52(5):525-30. doi: 10.3109/15563650.2014.913175. Epub 2014 May 5.

Evaluation of dexmedetomidine therapy for sedation in patients with toxicological events at an academic medical center.

Mohorn PL¹, Vakkalanka JP, Rushton W, Hardison L, Woloszyn A, Holstege C, Corbett SM.

Author information

Abstract

INTRODUCTION: Although clinical use of dexmedetomidine (DEX), an alpha2-adrenergic receptor agonist, has increased, its role in patients admitted to intensive care units secondary to toxicological sequelae has not been well established.

OBJECTIVES: The primary objective of this study was to describe clinical and adverse effects observed in poisoned patients receiving DEX for sedation.

METHODS: This was an observational case series with retrospective chart review of poisoned patients who received DEX for sedation at an academic medical center. The primary endpoint was incidence of adverse effects of DEX therapy including bradycardia, hypotension, seizures, and arrhythmias. For comparison, vital signs were collected hourly for the 5 h preceding the DEX therapy and every hour during DEX therapy until the therapy ended. Additional endpoints included therapy duration; time within target Richmond Agitation Sedation Score (RASS); and concomitant sedation, analgesia, and vasopressor requirements.

RESULTS: Twenty-two patients were included. Median initial and median DEX infusion rates were similar to the commonly used rates for sedation. Median heart rate was lower during the therapy (82 vs. 93 beats/minute, $p < 0.05$). Median systolic blood pressure before and during therapy was similar (111 vs. 109 mmHg, $p = 0.745$). Five patients experienced an adverse effect per study definitions during therapy. No additional adverse effects were noted. Median time within target RASS and duration of therapy was 6.5 and 44.5 h, respectively. Seventeen patients (77%) had concomitant use of other sedation and/or analgesia with four (23%) of these patients requiring additional agents after DEX initiation. Seven patients (32%) had concomitant vasopressor support with four (57%) of these patients requiring vasopressor support after DEX initiation.

CONCLUSION: Common adverse effects of DEX were noted in this study. The requirement for vasopressor support during therapy warrants further investigation into the safety of DEX in poisoned patients. Larger, comparative studies need to be performed before the use of DEX can be routinely recommended in poisoned patients.

PMID: 24792780 [PubMed - indexed for MEDLINE]

John R. McCarrey, Christopher Navara
University of Texas at San Antonio

PriStem – A Primate Resource for Developing Stem Cell Therapies

Stem cells and Regenerative Medicine offer a novel approach to the treatment of diseases and injuries that have otherwise been difficult to effectively treat with standard medical approaches. The advent of “induced pluripotent stem” cells (iPS cells) that can be derived from adult skin cells or blood cells offers an exciting alternative to the use of embryonic stem cells for this purpose, and the opportunity to develop “patient-specific” stem cells genetically and immunologically matched to each individual. Nevertheless, much research is still needed to optimize the efficacy and safety of stem cell-based therapeutic approaches before applying these to human patients. Nonhuman primates, such as baboons, are the most clinically relevant yet still experimentally manipulable animal model for studies of stem cell-based therapeutic approaches. We are developing “PriStem,” a comprehensive, user-friendly resource designed to provide nonhuman primate resources to foster the development, testing and optimization of therapeutic protocols involving patient-specific iPS cells prior to initiating tests in human patients. These resources include embryonic stem cells, induced pluripotent stem cells, and mesenchymal stem cells derived from baboons, plus access to matched living baboons from which each line of stem cells was derived. The Southwest National Primate Research Center in San Antonio is fully equipped to maintain living nonhuman primates and to perform experimental manipulations on these animals. PriStem is designed to function as a centralized, comprehensive resource that will complement work by researchers from many different institutions pursuing studies involving stem cells and their application to therapeutic treatments of diseases or debilitating conditions such as battlefield trauma. This resource is expected to accelerate the discovery, application and clinical translation of novel therapeutic approaches involving iPS and related types of stem cells.

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PNP GENERAL HOSPITAL
Camp Rafael Crame, Quezon City

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February 26, 2015

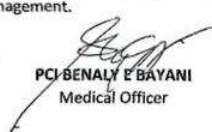
CLINICAL ABSTRACT

This is a case of **SENATOR JUAN PONCE ENRILE**, 91 Y/O, MALE, CIVILIAN, who was admitted at our institution since July 4, 2014. Patient is a diagnosed case of HCVD, Type 2 DM. However, on his course of admission at our institution, on February 25, 2015, patient developed productive cough of whitish to yellowish phlegm with an episode of hemoptysis associated with high grade fever of 1 day duration. On physical examination, patient is conscious, coherent, not in distress. BP 130/70, CAR 82, T 38.7, O2 sat 97%. Patient was noted to have flushed skin. On auscultation, rales was noted on the right lung base. Other PE findings were unremarkable.

**Diagnosis: Pneumonia in the elderly
To consider presumptive PTB**

Laboratory work-up and other diagnostic exam such as chest x-ray were requested and done with the following findings: CBC: hgb 128, hct 0.38, wbc 19.2, plt ct 248. Urinalysis: yellow, slightly turbid, negative sugar and proteins, rbc 3-5, pus cells 2-4. Chest x-ray initial reading: pneumonia right lower lobe with haziness on mid lobe probably PTB. Medical Management started with the following meds: Cefuroxime 750mg IV q8h ANST, Azithromycin 500mg tab OD, N-acetylcysteine 600mg tab OD and Paracetamol 300mg/amp, 1 amp IV q4h for Temp >38.5 or Paracetamol 500mg tab, 1 tab q4h for Temp >37.7.

In the event that the patient is an elderly, other diagnostic and laboratory exams such as blood culture, sputum GS CS, sputum AFB, ABG are considerably needed. Close monitoring is required as age of the patient is a significant risk factor for developing complications and deterioration of his physical condition, not to mention other co-morbidities such as hypertension and diabetes. With these foreseeable concerns and lack of facilities of PNPQH, the patient was transferred to tertiary hospital for further evaluation and management.


PCI BENALY E BAYANI
Medical Officer

NOTED:

PCI STANLEY G SEVILLANO

CLINICAL RESEARCH ARTICLE



Neonatal inflammation and its association with asthma and obesity in late childhood among individuals born extremely preterm

Crisma Emmanuel¹, Ali Oran², Elizabeth T. Jensen³, Raina N. Fichorova⁴, William A. Gower⁵, Eliana M. Perrin⁶, Keia Sanderson⁷, Andrew M. South^{3,8}, Sema Gogcu⁸, Jeffrey Shenberger⁹, Rachana Singh¹⁰, Kartikeya Makker⁶, Amanda L. Thompson¹¹, Hudson Santos¹², Rebecca C. Fry², T. M. O'Shea^{4,13} and for the ELGAN-ECHO Study Investigators*

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BACKGROUND: Asthma and obesity are frequent outcomes among individuals born extremely preterm and are associated with decreased lifespan. Neonatal inflammation is associated with chronic neurodevelopmental disorders; however, it is less studied in association with other later childhood chronic disorders in this population.

METHODS: Fourteen hospitals in 5 U.S. states enrolled 1506 infants born before 28 weeks of gestation in the Extremely Low Gestational Age Newborn cohort in 2004–2014. Neonatal blood spots were collected on postnatal days 1, 7, 14, 21, and 28, and used to measure 14 inflammation-related proteins. Associations were evaluated between high (top quartile) levels of proteins and two chronic health disorders at ages 10 and 15 years: physician-diagnosed asthma and obesity (body mass index ≥ 95 th percentile).

RESULTS: Few associations were found between high levels of 14 inflammation-related proteins, either on a single day or on multiple days, and either asthma or obesity. Similarly, few associations were found in analyses stratified by sex or presence/absence of prenatal inflammation.

CONCLUSIONS: In extremely preterm newborns, systemic elevations of inflammation-related proteins during the neonatal period were not associated with childhood asthma and obesity outcomes at 10 or 15 years of age.

Pediatric Research; <https://doi.org/10.1038/s41390-024-03325-x>

IMPACT:

- In the large multi-center Extremely Low Gestational Age Newborn (ELGAN) cohort, sustained elevation of neonatal levels of inflammation-related proteins was not consistently associated with asthma or obesity outcomes at 10 or 15 years of age.
- This finding contrasts with reported associations of perinatal inflammation with obesity at 2 years and neurodevelopmental disorders at 2–15 years in the ELGANs, suggesting that unlike neurodevelopment, peripubertal obesity and asthma may be driven by later childhood exposures.
- Future research on perinatal mechanisms of childhood asthma and obesity should account for both fetal and later exposures and pathways in addition to inflammation at birth.



Original Investigation | Public Health

Association of Neighborhood-Level Disadvantage With Alzheimer Disease Neuropathology

W. Ryan Powell, PhD; William R. Buckingham, PhD; Jamie L. Larson, PhD; Leigha Vilen, BS; Menggang Yu, PhD; M. Shahriar Salamat, MD, PhD; Barbara B. Bendlin, PhD; Robert A. Rissman, PhD; Amy J. H. Kind, MD, PhD

Abstract

IMPORTANCE Social determinants of health, such as income, education, housing quality, and employment, are associated with disparities in Alzheimer disease and health generally, yet these determinants are rarely incorporated within neuropathology research.

OBJECTIVE To establish the feasibility of linking neuropathology data to social determinants of health exposures using neighborhood disadvantage metrics (the validated Area Deprivation Index) and to evaluate the association between neighborhood disadvantage and Alzheimer disease–related neuropathology.

DESIGN, SETTING, AND PARTICIPANTS This cross-sectional study consisted of decedents with a known home address who donated their brains to 1 of 2 Alzheimer disease research center brain banks in California and Wisconsin between January 1, 1990, and December 31, 2016. Neither site had preexisting social metrics available for their decedents. Neuropathologic features were obtained from each site for data collected using the standardized Neuropathology Data Set form and from autopsy reports. Data were analyzed from June 7 to October 10, 2019.

EXPOSURES Geocoded decedent addresses linked to neighborhood disadvantage as measured by the Area Deprivation Index calculated for the year of death.

MAIN OUTCOMES AND MEASURES Presence of Alzheimer disease neuropathology. The association between neighborhood disadvantage and Alzheimer disease neuropathology was evaluated via logistic regression, adjusting for age, sex, and year of death.

RESULTS The sample consisted of 447 decedents (249 men [56%]; mean [SD] age, 80.3 [9.5] years; median year of death, 2011) spanning 24 years of donation. Fewer decedents ($n = 24$ [5.4%]) originated from the top 20% most disadvantaged neighborhood contexts. Increasing neighborhood disadvantage was associated with an 8.1% increase in the odds of Alzheimer disease neuropathology for every decile change on the Area Deprivation Index (adjusted odds ratio, 1.08; 95% CI, 1.07-1.09). As such, living in the most disadvantaged neighborhood decile was associated with a 2.18 increased odds of Alzheimer disease neuropathology (adjusted odds ratio, 2.18; 95% CI, 1.99-2.39).

CONCLUSIONS AND RELEVANCE The findings of this cross-sectional study suggest that social determinants of health data can be linked to preexisting autopsy samples as a means to study sociobiological mechanisms involved in neuropathology. This novel technique has the potential to be applied to any brain bank within the United States. To our knowledge, this is the first time Alzheimer disease neuropathology has been associated with neighborhood disadvantage.

Key Points

Question Can neighborhood disadvantage, a social determinant of health, be incorporated into existing brain bank data to evaluate the risk of biological outcomes, such as Alzheimer disease neuropathology?

Findings In this cross-sectional study using autopsy samples from 447 decedents, living in a disadvantaged neighborhood at the time of death was associated with an increased risk of presence of Alzheimer disease neuropathology when adjusting for age, sex, and year of death.

Meaning These findings suggest that neighborhood disadvantage can be geolinked to brain bank repositories and may be a marker for Alzheimer disease neuropathology.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.



OK, LET'S
BEGIN!

1. PURPOSE/BACKGROUND

BRIEF INTRODUCTION OF THE FIELD, HIGHLIGHTING THE QUESTIONS THAT LED TO THE STUDY.

WHY IS THIS STUDY IMPORTANT?

WHAT IS THE IMPACT?

Target your introduction to the understanding of a broad audience.

Don't use jargon or acronyms without defining what they mean.





2. RESEARCH QUESTION(S)

What is (are) your research question(s)?



What is your hypothesis?



What are the objectives you are trying to accomplish with this study?



What are the aims of the study?.

3. METHODS



List all methods used to answer the aims.



Be specific but succinct.



Detailed methodology should be reserved for the presentation itself.



Do not be too lengthy.

Specific questions may best be answered in presentation or published later with the full paper.

4. FINDINGS/RESULTS

- WHAT DATA WERE GENERATED IN RESPONSE TO THE QUESTIONS RAISED IN THE BACKGROUND?
- ONLY REPORT DATA YOU ARE CERTAIN OF – ACKNOWLEDGE IF THE STUDY IS INCOMPLETE
- TELL A FULL STORY - THE STRONGEST ABSTRACT IS THE ONE WITH THE MOST COMPLETE ANSWERS.





5. CONCLUSIONS

1

Put findings in the proper perspective.
Don't be far-reaching with the implications of the study findings.

2

Emphasize the strong points of your work: novelty, innovation and/or clinical relevance.

3

State how this work broadens the understanding of science in your discipline.



SCIENTIFIC ABSTRACTS USE AN “INTRODUCTION – BODY-CONCLUSION” STRUCTURE

Abstract Review Activity (10 min)

- Read the abstract.
- Form small groups of 2-4 with others that have read the same abstract.

Discuss:

- Does the abstract have the 5 items listed on this slide? Annotate!
- Reflect on good vs bad aspects of the abstract.
- Underline any jargon, acronyms, things that are unclear.



1. Purpose (Background)



2. Research Question(s)



3. Methods



4. Findings



5. Conclusions



Introduction

Body

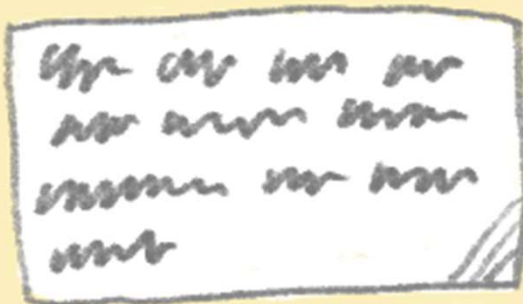
Conclusion



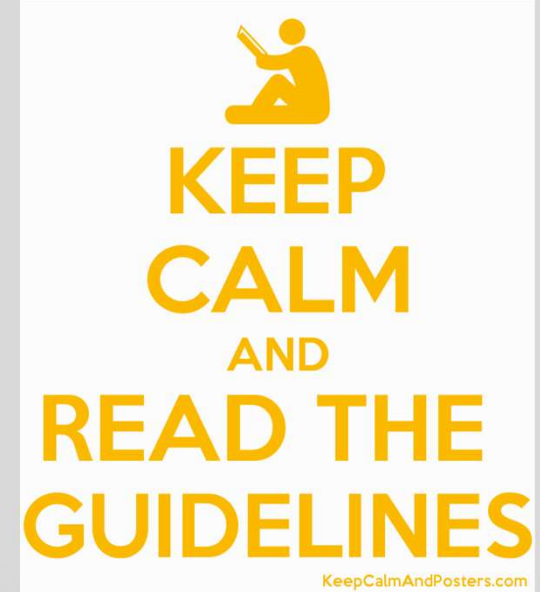
WHAT IF I
GET STUCK?

REMEMBER: MOST ABSTRACTS HAVE WORD OR
CHARACTER LIMITS!

NOTES



2 words
over the limit



**START BY WRITING FIRST...EVEN IF THE SENTENCES
DON'T MAKE SENSE!**



FIRST, WRITE FOR CONTENT.
THEN, WRITE FOR CLARITY.



EDIT, EDIT, EDIT!





TAKE IT STEP-BY-STEP

Write:

1. **1-2 introduction sentences** that explain topic, purpose, and research question(s).
2. **1-2 sentences describing your research methods** (this may also include the type of data analysis you used).
3. **1-2 sentences** describing the **results / findings**.
4. **1-2 sentences** containing your **conclusions and recommendations**.



REVISING THE ABSTRACT

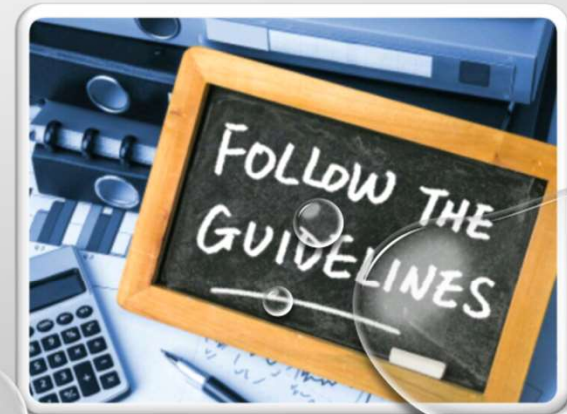
Read your abstract all the way through:

- **Add transition words** to tie ideas together.
- **Eliminate** unnecessary content and **add** things that are missing.
- Correct errors in mechanics, and **proofread**.

It's always easier to go back at the end and cut down words or find ways to reduce the word/character count.

FOLLOW THE GUIDELINES PRECISELY!

**OTHERWISE, YOUR ABSTRACT MAY BE
TRUNCATED OR PERHAPS REJECTED.**



GET APPROVAL FROM YOUR MENTOR AND CO-AUTHORS!

- DO **NOT** SUBMIT YOUR ABSTRACT WITHOUT FIRST GETTING APPROVAL FROM YOUR MENTOR.
 - YOUR MENTOR IS THE PRIMARY INVESTIGATOR ON THE STUDY, AND ULTIMATELY RESPONSIBLE FOR THE WORK.
- DO **NOT** SUBMIT YOUR ABSTRACT WITHOUT FIRST CONSULTING YOUR CO-AUTHORS.



IMPORTANT DATES:

Cardiovascular & Aging / TRAC Summer Research Symposium:



Abstract due –
Friday, July 11



Poster Title due –
Monday, July 14

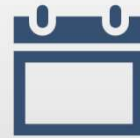


Posters due to Ms. Durr –
Thursday, July 17 at NOON



Research Symposium –
Thursday, July 24

BME/Informatics/eREU:



BME/eREU – bring abstract draft
to “Creating Value” workshop
Tuesday, July 15 at 9AM



Presentation Title due –
Weds, July 23 via CANVAS



Presentations due on CANVAS –
Tues, July 29 at 5PM



Research Symposium –
Thursday, July 31

